Solution Requirements

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
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1. Functional requirements

No.	Category	Requirement description	Priority
FR-1	Authentication	Users can sign up and sign in through Supabase Auth.	High
FR-2	Authorisation	A signed-in user can access only their own saved trips; ownership must be enforced through Row Level Security and backend checks.	High
FR-3	Trip input	Collect destination, dates / duration, interests, budget and travel pace.	High
FR-4	Validation	Reject missing or invalid trip details with a readable message.	High
FR-5	Itinerary generation	Frontend calls FastAPI, which uses Groq to generate an itinerary from the preferences.	High
FR-6	Presentation	Show itinerary information in a day-wise readable format.	High
FR-7	Persistence	Save trip data and list saved trips in My Trips.	High
FR-8	Edit	Restore saved preferences when a user edits an existing trip.	High
FR-9	Navigation	Preserve relevant trip state (e.g. destination) when moving between home, planner and itinerary.	Medium
FR-10	Error handling	Show actionable errors for failed API requests or AI-provider failures.	High

2. Non-functional requirements

Category	Requirement
Security	Keep secrets in server-side environment variables; use Supabase Row Level Security and least privilege; never expose the service-role or Groq key to the browser.
Reliability	Handle timeouts, an unavailable backend and invalid responses without crashing the UI.
Usability	Responsive, readable UI with clear labels, loading indicators and errors.
Maintainability	Separate React UI, API client, FastAPI routes, schemas and configuration.
Performance	Latency and error rate should be measured; no numeric targets are claimed without test evidence.
Compatibility	Support current desktop and mobile browsers.
Privacy	Collect only data needed for trip planning and account features; avoid logging secrets or tokens.

3. External interfaces

Vercel hosts the React/Vite frontend; Render hosts the FastAPI backend; Supabase provides authentication and PostgreSQL storage; Groq provides large-language-model inference for itinerary generation.